Assembly Instructions

Gas engine-generator sets

mtu 8V4000 GS – GG08V4000A1/A2 with engines 8V4000L32/ L32FB/L33 and 8V4000L64/L64FNER
mtu 12V4000 GS – GG12V4000A1/A2 with engines 12V4000L32/ L32FB/L33 and 12V4000L64/L64FNER
mtu 16V4000 GS – GG16V4000A1/A2 with engines 16V4000L32/ L32FB/L33 and 16V4000L64/L64FNER
mtu 20V4000 GS – GG20V4000A1/A2 with engines 20V4000L32/ L32FB/L33 and 20V4000L64/L64FNER

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A Rolls-Royce solution

Name	SLU name, model	Engine model	Application group
mtu 8V4000 GS	GG08V4000A1/A2	8V4000L32/L32FB/L33	3A, heavy duty, unrestricted, ICXN
	GG08V4000D1/D1M/D2	8V4000L64/L64FNER	
mtu 12V4000 GS	GG12V4000A1/A2	12V4000L32/L32FB/L33	
	GG12V4000D1/D1M/D2	12V4000L64/L64FNER	
mtu 16V4000 GS	GG16V4000A1/A2	16V4000L32/L32FB/L33	
	GG16V4000D1/D1M/D2	16V4000L64/L64FNER	
mtu 20V4000 GS	GG20V4000A1/A2	20V4000L32/L32FB/L33	
	GG20V4000D1/D1M/D2	20V4000L64/L64FNER	

Table 1: Applicability

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1 Preface

1.1 Preface

These Assembly Instructions are intended to assist project planners, system/component designers and installation personnel in planning or carrying out the installation of mtu components.

These Assembly Instructions also apply to gas engine-generator sets from Rolls-Royce Solutions.

The objective of these Assembly Instructions is to facilitate the assembly of components.

The Assembly Instructions do not relieve those responsible for the system from independently ensuring that installation is carried out in an appropriate manner.

In addition to the Assembly Instructions, the following documents included in the scope of supply must be observed:
- Planning drawing
- Schematic diagrams
- TEN data
- Arrangement drawings
- Operating instructions (see business portal)
- Maintenance schedule (see business portal)
- Fluids and lubricants specifications (see business portal)
- mtu standards

Each document is taken from the respective specifications document required for the order.

Adherence to the prescribed schedule for maintenance tasks will have a positive effect on operational safety, reliability and long service life. During the planning or installation of the plant system, ease of access for operation, maintenance and repair work is to be ensured.

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2 Safety

2.1 Important provisions for all products

General

This product may pose a risk of injury or damage in the following cases:
- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number. This data must match the specifications in these instructions.

Nameplates, model designation or serial number can be found on the product.

All EU-certified engines delivered by Rolls-Royce Solutions come with a second nameplate. This second nameplate is delivered "loosely" with the engine. If the nameplate secured to the engine after installation in the vehicle/system is not visible without the removal of components, the system integrator must install the second nameplate in a clearly visible area on the vehicle/system.

Emissions regulations

Prior to operation, make sure that the latest version is used. The latest version can be found at:http://www.mtu-solutions.com

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2.2 Intended use of gas systems

Intended use

The machine is intended to be used exclusively for electrical power generation and waste heat utilization in stationary operation, either in sufficiently ventilated rooms or in container solutions.

A gas/air mixture is fed to the internal combustion reciprocating piston engine. This is converted into mechanical work in the individual cylinders through combustion. The flywheel of the engine drives the synchronous generator connected via an elastic coupling, resulting in a conversion to electric energy. The engine and the generator are joined to the base frame with resilient, vibration-damped elements.

For mains frequency 60 Hz, depending on the series either a gearbox is used between the engine and generator or the engine speed is increased accordingly. The waste heat generated during the combustion process is dissipated to the surroundings via coolers (power generator) and/or made available to the customer via heat exchangers (combined heat and power plant).

Area of application	Applicable	Comment
Industry	Yes	–
Business/Service	Yes	In some individual cases, however, such as with swimming pool facilities, schools, heating stations for apartment buildings and the like, no end user in the classic sense is involved
Household	No	Operation is excluded
Potentially explosive area	No	Operation is not permitted in potentially explosive areas
Underground	No	Operation is excluded
Mobile use	No	Operation is excluded

Table 2: Area of application of machine

There are 3 operating modes of the machine:
- Mains parallel operation
- Mains failure operation
- Isolated operation

Foreseeable misuse

Any other use, particularly misuse or wrong parametrization, is considered as being contrary to the intended purpose. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

Operation

Operation is not permissible if the physical conditions specified on the technical data sheet or on/in other documents (concerning temperature, pressure, site altitude, etc.) are not complied with. Operation is not permissible without protective equipment, usage of approved spare parts, supplied accessory parts, exclusion of conversions, warning of foreseeable misuse.

Operation is only permissible with the use of fluids and lubricants (natural gas, biogas, quality-compliant gas, quality-compliant water, lubricants) in accordance with the (→ Fluids and Lubricants Specifications).

Operation is only permissible with preservation agents approved by Rolls-Royce Solutions in accordance with the (→ Preservation and Represervation Specifications).

Operation is only permissible after proper integration/installation.

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The specifications of the manufacturer will be amended or supplemented as necessary. Prior to operation, make sure that the latest version is used. The latest version can be found at:
- http://www.mtu-solutions.com

Surroundings, climate, temperature

The weather plays no role with regard to the intended use of the engine-generator set/module provided that operation takes place inside an installation room. For container applications, however, the permissible outdoor temperatures at the installation site play a decisive role. The product-specific ambient conditions such as climate and temperature are indicated in the technical description.

Prerequisite for proper use

Fulfillment of all the requirements set out in the “Installation conditions” document.

Any use that does not fall within the described limits for use is regarded as improper.

Work which requires communication with the control units must only be carried out with a diagnosis software or with tools approved by the manufacturer.

Modifications or conversions

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.

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2.3 Personnel and organizational requirements

Organizational measures of the user/manufacturer

This manual must be issued to all personnel involved in operation, maintenance, repair, assembly, installation, or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair, assembly, installation, and transport personnel at all times.

Personnel must receive instruction on product handling and repair based on this manual. In particular, personnel must have read and understood the safety requirements and warnings before starting work.

This is important in the case of personnel who only occasionally perform work on or around the product. Such personnel must be instructed repeatedly.

Personnel requirements

All work on the product must be carried out by trained, instructed and qualified personnel only:
- Training at the Training Center of the manufacturer
- Qualified personnel from the areas mechanical engineering, plant construction, and electrical engineering and also for work with live parts

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair, assembly, installation, and transport in writing.

Personnel shall not report for duty under the influence of alcohol, drugs or strong medication.

Clothing and personal protective equipment

Always wear appropriate personal protective equipment, e.g. safety shoes, ear protectors, protective gloves, goggles, breathing mask. Follow the instructions concerning personal protective equipment in the respective descriptions of the individual activities or the manufacturer's documentation included in the delivery.

When working on live components, in particular on electrical switchgear or on high-voltage component, wear special protective clothing according to IEC 61482.

Safe handling of Substances of Very High Concern pursuant to the REACH regulation (Registration, Evaluation, Authorization and restriction of CHemicals): We recommend wearing protective gloves at all times in order to reduce risk when working.

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2.4 Safety regulations for commissioning and operation

Safety regulations for commissioning

Install the product correctly and carry out acceptance in accordance with the manufacturer's specifications before commissioning the product. All necessary approvals must be granted by the relevant authorities and all requirements for commissioning must be fulfilled.

Whenever the product is subsequently put into operation ensure that:
- All personnel is clear of the danger zone surrounding moving parts of the machine. Electrically-actuated linkages may be set in motion when the Engine Control Unit (governor) is switched on.
- All maintenance and repair work has been completed.
- All loose parts have been removed from rotating machine components.
- All protective devices are in place.
- All lines bearing media are ready for operation.
- The walkways in the operating room are unobstructed and safe.
- No persons wearing pacemakers or any other technical body aids are present.
- Adequate ventilation of the operating room is ensured for any operating status.
- In the first few hours of operation, the product emits gases as a result of smoldering e.g. lacquers or oil. These gases may be hazardous to health. Always wear respiratory protection in the operating room during this period.
- The exhaust system is leak-tight and the gases are vented to atmosphere.
- The product is free of any damage, this applies in particular to lines and cabling.
- All components are properly grounded. Ground separately by means of a grounding stake as necessary.
- The power supply (mains plug) is always freely accessible to facilitate disconnecting the device or system from the power supply at any time.
- Battery terminals, generator terminals or cables are protected against accidental contact.
- All connections have been correctly allocated e.g. +/- polarity, fuel line/reducing agent line, supply/return.

Immediately after putting the product into operation, make sure that all control and display instruments as well as the monitoring, signaling and alarm systems work properly.

Smoking is prohibited in the area of the product.

Safety regulations during operation

The operator must be familiar with the control and display elements.

The operator must be familiar with the consequences of any operations performed.

During operation, the display instruments and monitoring units must be observed with regard to present operating status, violation of limit values and warning or alarm messages.

Malfunctions and emergency stop

Practice emergency procedures, especially emergency stopping, at regular intervals.

Take the following steps if any system malfunctions are detected or signaled by the system:
- Inform supervisor(s) in charge.
- Analyze the message.
- Respond by taking any necessary emergency action, e.g. emergency stop.

After a safety shutdown, the engine may only be started after the cause of the shutdown has been eliminated.

Contact Service if the root cause of the malfunction cannot be clearly identified.

Operation

Do not remain in the operating room when the product is running unless absolutely necessary. Keep your stay as short as possible.

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